

DIGITAL LOGIC NOTES

Unit 1: Number Systems and Codes

1. Decimal, Binary, Octal, and Hexadecimal number systems.
2. Conversion between number systems.
3. Binary arithmetic (addition, subtraction using 2's complement).
4. BCD, Gray code, ASCII.

Unit 2: Logic Gates and Boolean Algebra

1. Basic Logic Gates: AND, OR, NOT.
2. Universal Gates: NAND and NOR.
3. Boolean Laws and Theorems.
4. De Morgan's Theorem.
5. SOP and POS forms.

Unit 3: K-Map and Logic Simplification

1. Minterms and Maxterms.
2. Karnaugh Map (K-Map) simplification (2, 3, 4 variables).
3. Simplification using K-Map.

Unit 4: Combinational Logic Circuits

1. Half Adder and Full Adder.
2. Half Subtractor and Full Subtractor.
3. Multiplexers and Demultiplexers.
4. Encoders and Decoders.

Unit 5: Sequential Logic Circuits

1. Flip-Flops: SR, JK, D, T.
2. Truth tables and characteristic equations.
3. Registers and Counters.
4. Shift Registers.

Unit 6: Memory and Logic Families

1. RAM and ROM.
2. PROM, EPROM, EEPROM.
3. TTL and CMOS logic families.